higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 4

(12010484)

13 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. ALL drawings must be neat and well balanced.
7. Write neatly and legibly.

Copyright reserved

QUESTION 1

1.1 Complete the following sentences by filling in the missing word or words. Write only the word or words next to the question number (1.1.1–1.1.6) in the ANSWER BOOK.

1.1.1 Plastics may be divided into two categories, namely thermoplastic and … plastics.

1.1.2 Thermoplastics soften when … and harden when cooled.

1.1.3 In scientific terms, plastics are also known as …

1.1.4 In construction, elastomers are used mainly as … and sealants.

1.1.5 Natural rubber is an elastomer that originally came from the milky … of rubber tree.

1.1.6 Solvent welding can be used to join and seal rigid … pipes.

(6 × 1) (6)

1.2 Explain FOUR procedures for applying silicon sealant to masonry and concrete. (4)

1.3 Explain how you will carry out the knife slither test. (3)

1.4 There are certain advantages in using composite material on a building site.

Name THREE types of composite materials. (3)

1.5 State FOUR advantages of using composite material. (4)

[20]

QUESTION 2

2.1 Cured natural rubber can and be used for various purposes.

State FIVE uses of natural rubber in the construction industry. (5)

2.2 Describe a primer and explain its application when preparing the reaming of a saw-cut construction joint in a concrete slab. (2)

2.3 You have been called out to a site where you observe that the acid-based silicone sealant did not adhere to the material it was supposed to seal.

State THREE possible reasons why the sealant did not adhere to the material. (3)

[10]

Copyright reserved

QUESTION 3

3.1 State SIX limitations and weaknesses of cyanoacrylate adhesives. (6)

3.2 You have to bond two wooden surfaces by using a contact adhesive.

Briefly explain the steps to perform this task successfully. (7)

3.3 Epoxies have very high strength bonding capabilities, but also limitations in respect of the building industry.

State TWO limitations of epoxies. (2)
[15]

QUESTION 4

4.1 Lead is a heavy metal that can cause serious health problems unless one takes suitable precautions.

List FIVE conditions, detrimental to your health that long-term exposure to lead paint can cause in adults. (5)

4.2 Explain how you will paint a kitchen surface consisting of unfinished (new) wood. (8)

4.3 List THREE coat paint systems. (3)

4.4 How would you etch and prime a new galvanised surface? (4)
[20]

QUESTION 5

5.1 Define the term *efflorescence.* (2)

5.2 Water repellents penetrate the surface of a structure to between 3–5 mm.

Name THREE types of water repellents. (3)

5.3 Explain the meaning of the following terms:

5.3.1 Footings

5.3.2 DPC

5.3.3 Weep-holes

5.3.4 Flashing

5.3.5 Cavity

(5 × 2) (10)
[15]

Copyright reserved

QUESTION 6

6.1 State FOUR precautionary procedures to reduce the risk of fire, skin burns and respiratory disease. (4)

6.2 Name FOUR generic advantages of asphalt. (4)

6.3 Identify the following road-construction plant images below. Write ONLY the answer and the question number (6.3.1–6.3.2) in the ANSWER BOOK.

6.3.1



6.3.2



(2 × 1) (2)

[10]

QUESTION 7

7.1 Identify the following fasteners.

7.1.1



7.1.2



7.1.3



(3 × 1) (3)

7.2 State FIVE functions of the bolt list. (3)

7.3 State FOUR factors one should consider when torque tensioning on a structure. (4)

[10]

TOTAL 100

Copyright reserved